

24-51-Q3

Q: (a) F L D

(b) SUM

Solution (a) ① within - class scatter

$$S_w = S_1 + S_2 = \begin{bmatrix} 223.107 & -13.506 \\ -13.506 & 177.306 \end{bmatrix}$$

$$m_1 - m_2 = \begin{bmatrix} 2.444 \\ -2.595 \end{bmatrix}$$

$$m_1 + m_2 = \begin{bmatrix} -0.718 \\ -0.593 \end{bmatrix}$$

$$w = S_w^{-1} (m_1 - m_2) = \begin{bmatrix} 0.01012 \\ -0.01387 \end{bmatrix}$$

$$w_0 = - \frac{w^T (m_1 + m_2)}{2} = -0.0004797$$

$$\text{So } y(x) = w^T x + w_0$$

$$= \begin{bmatrix} 0.01012 & -0.01387 \end{bmatrix} x - 0.0004797$$

$$x_2 = \begin{bmatrix} -0.176 \\ 0.224 \end{bmatrix}$$

$$y(x_2) = \begin{bmatrix} 0.01012 & -0.01387 \end{bmatrix} \begin{bmatrix} -0.176 \\ 0.224 \end{bmatrix} - 0.0004797$$

$$= -0.0053677 < 0$$

assign x_2 to class 2

(b) ① SUM w^* calculate

$$w^* = \sum_{i=1} \alpha(i) d(i) x(i)$$

$$\begin{aligned}
 &= 0.3784 \times 1 \times [0.287 \ 0.0421 \ 0.821]^T \\
 &+ 0.051 \times 1 \times [4.32 \ -0.572 \ 3.45]^T \\
 &+ 0.0505 \times (-1) \times [3.922 \ -1.421 \ -2.721]^T \\
 &+ 0.3789 \times (-1) \times [0.324 \ -2.234 \ 1.122]^T \\
 &= [0.008096 \ 0.9050 \ 0.1989]^T
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad w_o &= \frac{1}{N_s} \sum_{\alpha(i) \neq 0} [d(i) - (w^*)^T x(i)] \\
 &= \frac{1}{4} \times \left(\left(1 - [0.008096 \ 0.9050 \ 0.1989] \begin{bmatrix} 0.287 \\ 0.0421 \\ 0.821 \end{bmatrix} \right) \right.
 \end{aligned}$$

$$\begin{aligned}
 &+ \left(1 - [0.008096 \ 0.9050 \ 0.1989] \times [4.32 \ -0.572 \ 3.45]^T \right) \\
 &+ \left(-1 - [0.008096 \ 0.9050 \ 0.1989] \times [3.922 \ -1.421 \ -2.721]^T \right) \\
 &+ \left(-1 - [0.008096 \ 0.9050 \ 0.1989] \times [0.324 \ -2.234 \ 1.122]^T \right) \\
 &= \frac{1}{4} \times [(1 - 0.2037) + (1 - 0.2035) + (-1 + 1.7955) + (-1 + 1.7966)]
 \end{aligned}$$

$$= 0.7961$$

③ classifier

$$y = [0.008096 \quad 0.9050 \quad 0.1989] x + 0.7961$$

④ test sample $x_3 = [0.225 \quad 2.252 \quad -2.613]^T$

$$y = 2.2920 > 0$$

So we classify x_3 into class 1 (label 1)